

ASSESSMENT TOOL 3- CORRELATION AND PATTERN ANALYSIS

Course Code: DSA0504	Course Title: Query Processing for Data Science
CO Assessed: CO4	Assessment: Tool 3– Correlation and pattern Analysis
Weightage: 25	Total Marks: 25 Marks
CO4 Statement: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.	
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Correlation vs Causation – CGPA and Salary Package Analysis

1. Problem Statement

A university placement cell analyzes data from 1,500 final-year students to understand the relationship between students' academic performance, internship experience, skills, and salary packages. The university observes that students with higher CGPA generally receive better salary packages and that students with higher CGPA are also more likely to have internship experience. Based on these observations, the university concludes that improving CGPA alone will directly increase students' salary packages. The objective of this analysis is to determine whether this conclusion is valid and to identify other factors that may influence salary packages.

2. Correlation Between CGPA and Salary Package

A positive correlation may exist between CGPA and salary package. This means that, in the dataset, students with higher CGPA tend to receive higher salary packages. However, this relationship represents only an association. It does not prove that increasing CGPA directly causes an increase in salary.

Interpretation:

Higher CGPA → Usually associated with higher salary, but causation is not established.

3. Correlation Between Internship Experience and Salary Package

Students with internship experience may receive better salary packages because internships provide:

- Practical industry experience
- Technical knowledge
- Exposure to real-world projects
- Better problem-solving skills
- Improved interview performance

Therefore, internship experience can also be positively related to salary packages.

4. Correlation Between CGPA and Internship Experience

The data also indicates that students with higher CGPA are more likely to have internship experience.

This creates an important analytical issue. A student may receive a higher salary not only because of CGPA but also because of the additional experience obtained through internships.

Therefore, internship experience can act as a confounding variable in the relationship between CGPA and salary.

5. Correlation vs Causation

Correlation means that two variables change together.

For example:

Higher CGPA $\leftrightarrow$ Higher Salary

But causation means that a change in one variable directly produces a change in another variable.

Therefore:

Higher CGPA $\rightarrow$ Higher Salary

cannot be concluded from correlation alone.

Even if a strong positive correlation is obtained, it does not prove that increasing a student's CGPA will automatically increase their salary package.

6. Confounding Variables

Several hidden or confounding variables may influence the observed relationship between CGPA and salary.

Internship Experience

Students with internships may have better practical knowledge and may perform better during placements.

Technical Skills

Programming, data analysis, AI/ML, problem-solving, and other technical skills can significantly influence job opportunities and salary.

Projects and Certifications

Students with strong projects and professional certifications may receive better job offers.

Communication and Interview Skills

A student with good communication, aptitude, group discussion, and interview skills may receive a higher package even with a moderate CGPA.

Department or Branch

Salary opportunities may differ between departments such as CSE, AI & DS, ECE, Mechanical, and Civil Engineering.

Thus, the observed salary difference cannot be attributed to CGPA alone.

7. Critical Evaluation of the University's Conclusion

The university's statement that:

“Improving CGPA alone will directly increase students' salary packages.”

is not statistically justified.

CGPA may be an important factor considered by recruiters, but salary packages depend on several other variables.

For example, consider two students:

Student	CGPA	Internship	Technical Skills	Salary
A	9.0	No	Average	Moderate
B	8.0	Yes	Excellent	High

Student B may receive a higher package despite having a lower CGPA because of better practical experience and technical skills.

Therefore, CGPA alone cannot explain salary differences.

8. Analytical Strategy

To distinguish genuine predictive relationships from misleading correlations, the following analysis should be performed.

Step 1 – Correlation Analysis

Calculate correlations among:

- CGPA and Salary

- Internship Experience and Salary
- Technical Skills and Salary
- Projects and Salary
- CGPA and Internship Experience

A correlation matrix can be used to compare these relationships.

Step 2 – Data Visualization

Create a scatter plot of CGPA vs. Salary Package.

This helps identify:

- Positive or negative trends
- Outliers
- Clusters of students
- Non-linear relationships

Step 3 – Subgroup Analysis

Students should be divided into groups such as:

- Students with internships
- Students without internships
- Different departments
- Different technical-skill levels

The relationship between CGPA and salary should then be examined separately within each group.

This helps determine whether the overall correlation is hiding important subgroup differences.

Step 4 – Multiple Linear Regression

Use salary package as the dependent variable.

Possible independent variables are:

Salary = $f(\text{CGPA, Internship, Technical Skills, Projects, Department, Communication Skills})$

Multiple regression helps estimate the effect of CGPA while statistically controlling for other variables.

Step 5 – Compare Models

Two models can be created:

Model 1:

Salary = CGPA

Model 2:

Salary = CGPA + Internship + Technical Skills + Projects + Other Factors

If Model 2 explains salary substantially better than Model 1, it indicates that salary depends on several factors rather than CGPA alone.

9. Code

```
# =====  
# CORRELATION VS CAUSATION:  
# CGPA AND SALARY PACKAGE ANALYSIS  
# =====  
  
# =====  
# 1. CREATE DATASET  
# =====  
  
# Set seed for reproducibility  
np.random.seed(42)  
  
# Number of students  
n = 1500  
  
# Generate CGPA values  
cgpa = np.clip(  
    np.random.normal(7.5, 0.9, n),  
    5.0,  
    10.0  
)  
  
# Generate internship status  
internship = np.random.binomial(  
    1,  
    np.clip(0.25 + 0.10 * (cgpa - 5), 0.10, 0.85),  
    n  
)  
  
# Generate technical skills score  
technical_skills = np.clip(  
    45 + 5.5 * (cgpa - 5) + np.random.normal(0, 10, n),  
    20,  
    100  
)  
  
# Generate number of projects  
projects = np.clip(  
    np.round(  
        1.5 + 0.45 * (cgpa - 5) +  
        np.random.normal(0, 1.2, n)
```

```

    ),
    0,
    6
).astype(int)

# Generate communication skills score
communication_skills = np.clip(
    55 + 4 * (cgpa - 5) +
    np.random.normal(0, 12, n),
    20,
    100
)

# Generate department
departments = np.random.choice(
    ["CSE", "AI & DS", "ECE", "Mechanical", "Civil"],
    size=n,
    p=[0.30, 0.20, 0.20, 0.15, 0.15]
)

# Department effect on salary
department_effect = {
    "CSE": 2.0,
    "AI & DS": 2.5,
    "ECE": 1.0,
    "Mechanical": 0.0,
    "Civil": -0.5
}

# Generate salary package in LPA
salary = (
    2.5
    + 0.55 * cgpa
    + 2.0 * internship
    + 0.035 * technical_skills
    + 0.35 * projects
    + 0.02 * communication_skills
    + np.array([department_effect[d] for d in departments])
    + np.random.normal(0, 1.5, n)
)

# Minimum salary = 2 LPA
salary = np.clip(salary, 2, None)

# =====
# 2. CREATE DATAFRAME
# =====

df = pd.DataFrame({
    "CGPA": cgpa,
    "Internship": internship,
    "Technical_Skills": technical_skills,
    "Projects": projects,
    "Communication_Skills": communication_skills,
    "Department": departments,
    "Salary_Package": salary
})

```

```

# Convert internship values into Yes/No
df["Internship_Status"] = df["Internship"].map({
    0: "No",
    1: "Yes"
})

# =====
# 4. CORRELATION ANALYSIS
# =====

numeric_columns = [
    "CGPA",
    "Internship",
    "Technical_Skills",
    "Projects",
    "Communication_Skills",
    "Salary_Package"
]

correlation_matrix = df[numeric_columns].corr()

print("\n" + "=" * 60)
print("CORRELATION MATRIX")
print("=" * 60)

print(correlation_matrix.round(3))

# =====
# 5. CORRELATION HEATMAP
# =====

plt.figure(figsize=(10, 7))

sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap="coolwarm",
    fmt=".2f",
    linewidths=0.5
)

plt.title("Correlation Matrix - Student Placement Data")
plt.tight_layout()
plt.show()

# =====
# 6. CGPA VS SALARY SCATTER PLOT
# =====

plt.figure(figsize=(9, 6))

sns.regplot(
    data=df,
    x="CGPA",
    y="Salary_Package",

```



```

        scatter_kws={"alpha": 0.35},
        line_kws={"color": "red"}
    )

plt.title("CGPA vs Salary Package")
plt.xlabel("CGPA")
plt.ylabel("Salary Package (LPA)")
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

# =====
# 7. SUBGROUP ANALYSIS BY INTERNSHIP
# =====

group_correlations = df.groupby(
    "Internship_Status"
).apply(
    lambda group:
        group["CGPA"].corr(group["Salary_Package"])
)

print("\n" + "=" * 60)
print("CGPA-SALARY CORRELATION BY INTERNSHIP STATUS")
print("=" * 60)

print(group_correlations.round(3))

# =====
# 8. CGPA VS SALARY BY INTERNSHIP STATUS
# =====

plt.figure(figsize=(9, 6))

sns.scatterplot(
    data=df,
    x="CGPA",
    y="Salary_Package",
    hue="Internship_Status",
    alpha=0.5
)

plt.title("CGPA vs Salary by Internship Status")
plt.xlabel("CGPA")
plt.ylabel("Salary Package (LPA)")
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

# =====
# 9. SIMPLE LINEAR REGRESSION
#     Salary = CGPA
# =====

X_simple = df[["CGPA"]]

```

```

y = df["Salary_Package"]

X_train, X_test, y_train, y_test = train_test_split(
    X_simple,
    y,
    test_size=0.20,
    random_state=42
)

simple_model = LinearRegression()

simple_model.fit(
    X_train,
    y_train
)

simple_predictions = simple_model.predict(X_test)

simple_r2 = r2_score(
    y_test,
    simple_predictions
)

simple_rmse = np.sqrt(
    mean_squared_error(
        y_test,
        simple_predictions
    )
)

print("\n" + "=" * 60)
print("SIMPLE LINEAR REGRESSION")
print("=" * 60)

print(
    "Intercept:",
    round(simple_model.intercept_, 3)
)

print(
    "CGPA Coefficient:",
    round(simple_model.coef_[0], 3)
)

print(
    "R-squared:",
    round(simple_r2, 3)
)

print(
    "RMSE:",
    round(simple_rmse, 3)
)

# =====
# 10. MULTIPLE LINEAR REGRESSION
# =====

```

```

features = [
    "CGPA",
    "Internship",
    "Technical_Skills",
    "Projects",
    "Communication_Skills",
    "Department"
]

X = df[features]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

# Numeric features
numeric_features = [
    "CGPA",
    "Internship",
    "Technical_Skills",
    "Projects",
    "Communication_Skills"
]

# Categorical feature
categorical_features = [
    "Department"
]

# Preprocessor
preprocessor = ColumnTransformer(
    transformers=[
        (
            "numeric",
            "passthrough",
            numeric_features
        ),
        (
            "categorical",
            OneHotEncoder(
                drop="first",
                handle_unknown="ignore"
            ),
            categorical_features
        )
    ]
)

# Create pipeline
multiple_model = Pipeline(
    steps=[
        (
            "preprocessor",
            preprocessor

```

```

        ),
        (
            "regressor",
            LinearRegression()
        )
    ]
)

# Train model
multiple_model.fit(
    X_train,
    y_train
)

# Predict
multiple_predictions = multiple_model.predict(
    X_test
)

# Calculate performance
multiple_r2 = r2_score(
    y_test,
    multiple_predictions
)

multiple_rmse = np.sqrt(
    mean_squared_error(
        y_test,
        multiple_predictions
    )
)

print("\n" + "=" * 60)
print("MULTIPLE LINEAR REGRESSION")
print("=" * 60)

print(
    "R-squared:",
    round(multiple_r2, 3)
)

print(
    "RMSE:",
    round(multiple_rmse, 3)
)

# =====
# 11. MODEL COMPARISON
# =====

model_comparison = pd.DataFrame({
    "Model": [
        "Simple Regression - CGPA Only",
        "Multiple Regression - All Factors"
    ],
    "R_squared": [
        simple_r2,

```

```

        multiple_r2
    ],
    "RMSE": [
        simple_rmse,
        multiple_rmse
    ]
})

print("\n" + "=" * 60)
print("MODEL COMPARISON")
print("=" * 60)

print(
    model_comparison.round(3).to_string(index=False)
)

# =====
# 12. ACTUAL VS PREDICTED SALARY
# =====

plt.figure(figsize=(8, 6))

sns.scatterplot(
    x=y_test,
    y=multiple_predictions,
    alpha=0.6
)

minimum = min(
    y_test.min(),
    multiple_predictions.min()
)

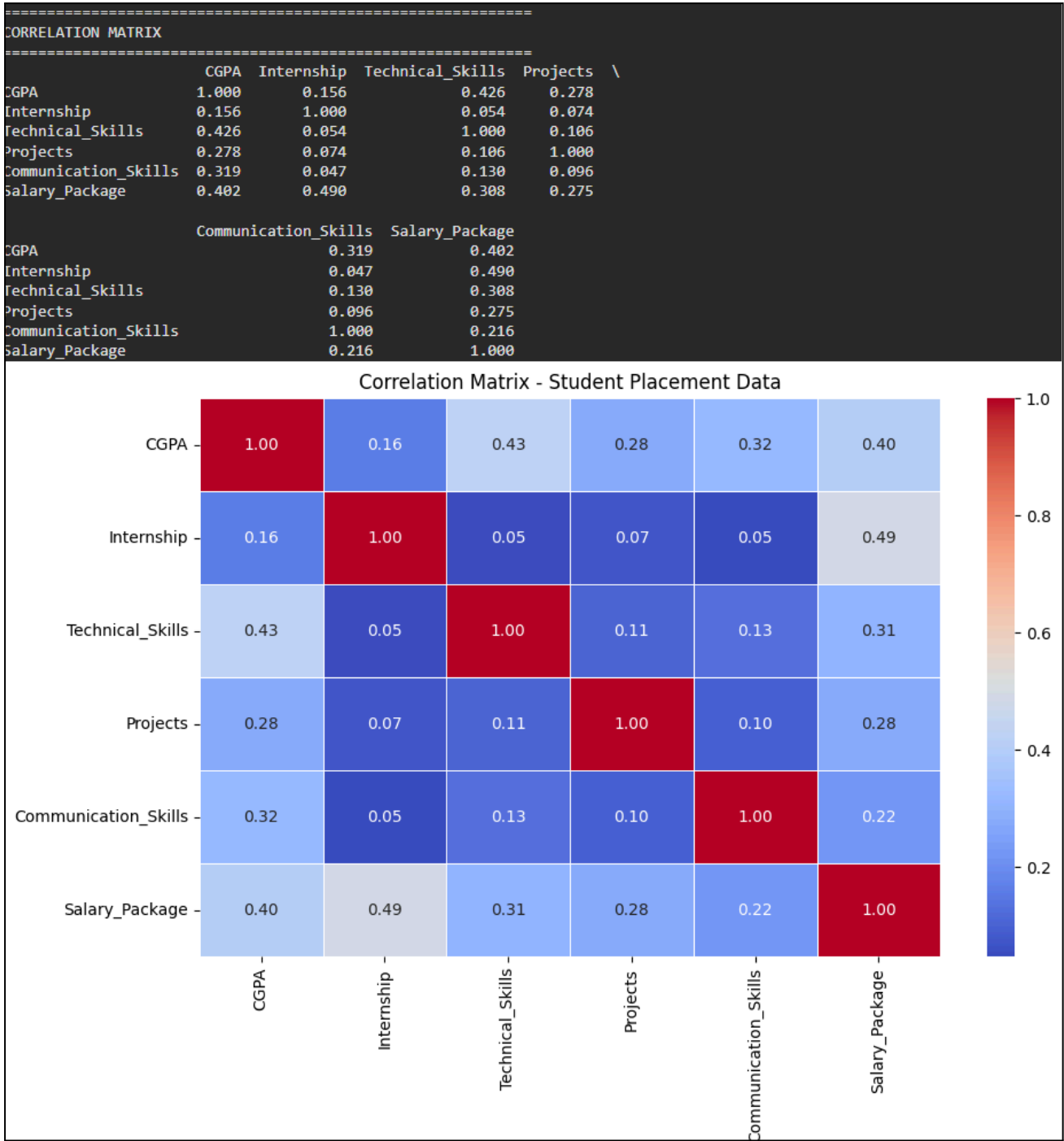
maximum = max(
    y_test.max(),
    multiple_predictions.max()
)

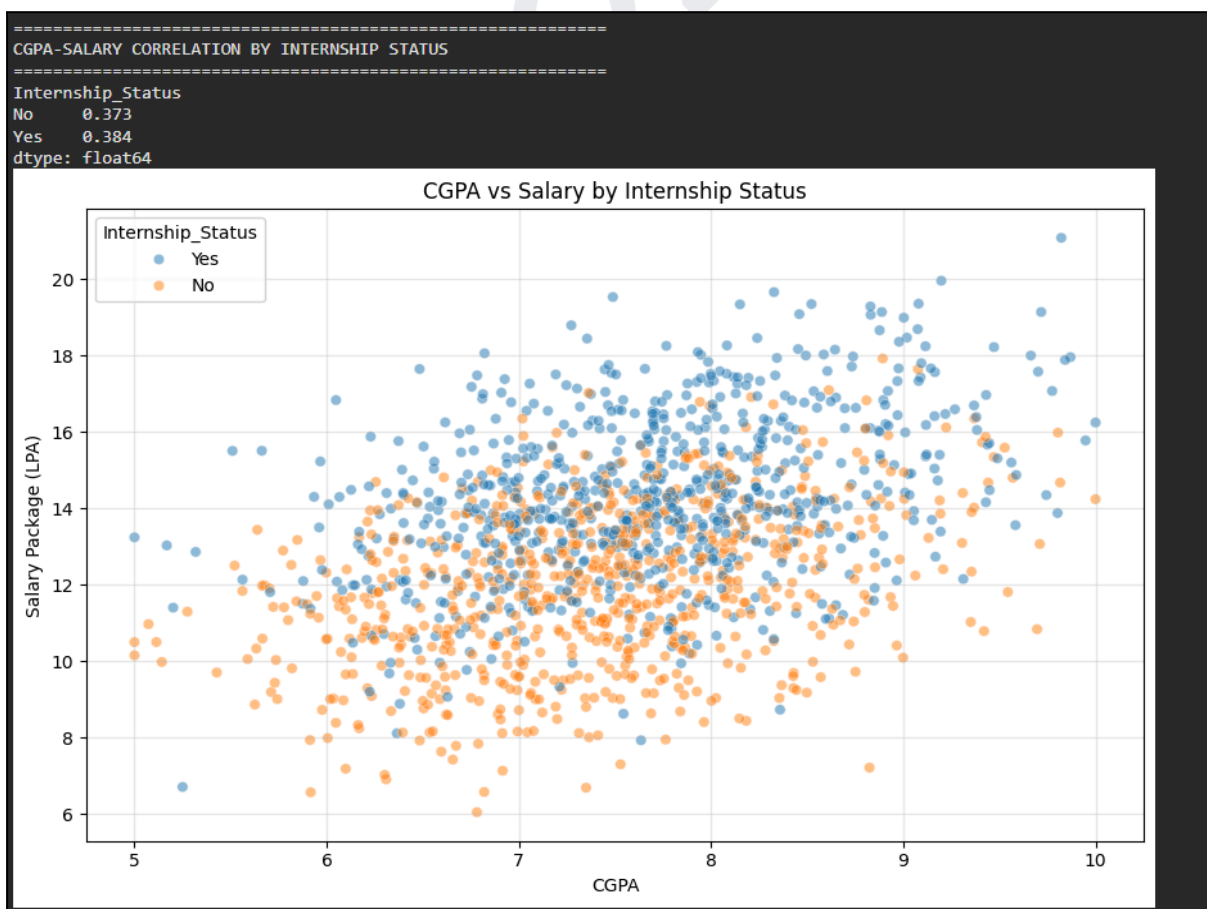
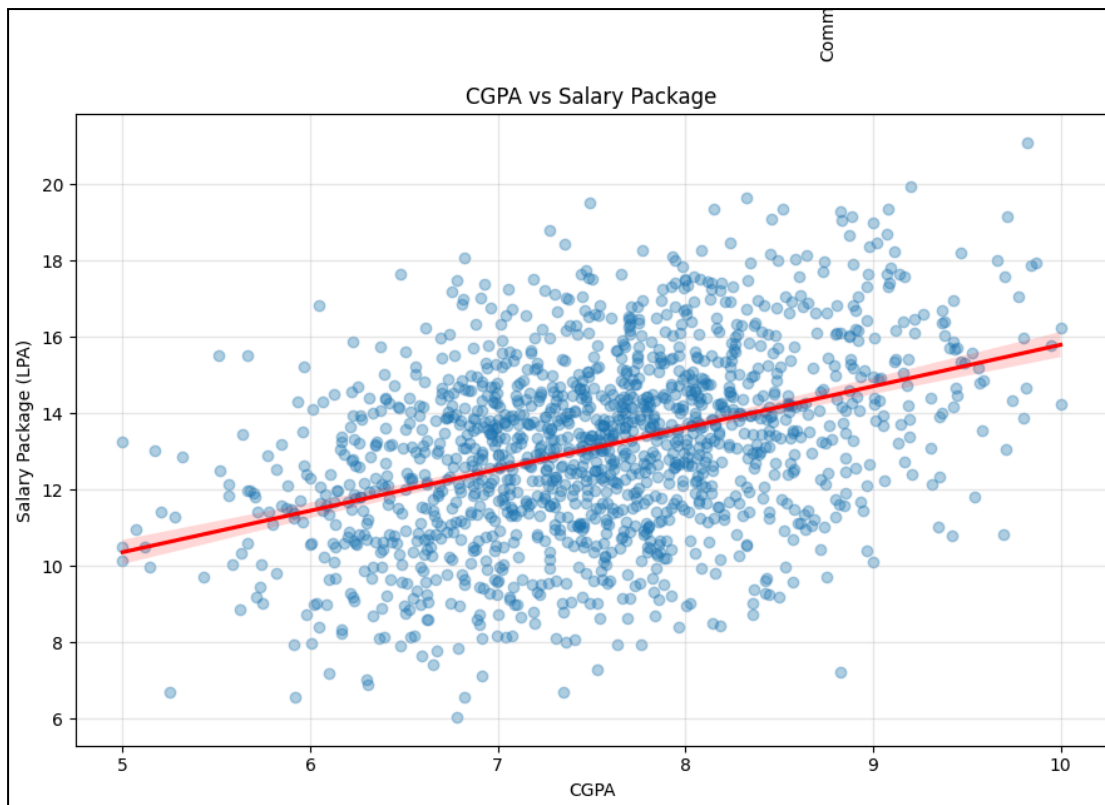
plt.plot(
    [minimum, maximum],
    [minimum, maximum],
    color="red",
    linestyle="--"
)

plt.title("Actual vs Predicted Salary Package")
plt.xlabel("Actual Salary Package (LPA)")
plt.ylabel("Predicted Salary Package (LPA)")
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

```

10. Output





SIMPLE LINEAR REGRESSION

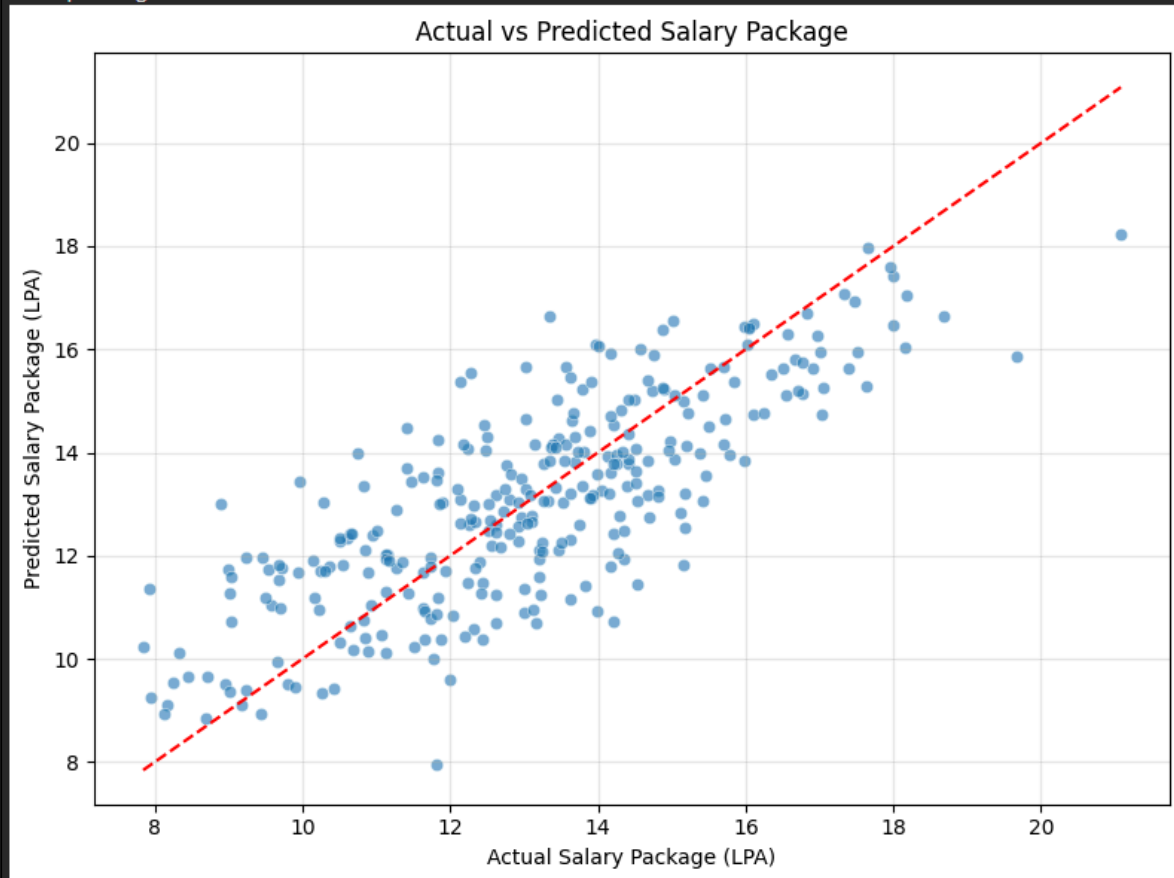
Intercept: 5.17
CGPA Coefficient: 1.056
R-squared: 0.204
RMSE: 2.117

MULTIPLE LINEAR REGRESSION

R-squared: 0.617
RMSE: 1.468

MODEL COMPARISON

Model	R_squared	RMSE
Simple Regression - CGPA Only	0.204	2.117
Multiple Regression - All Factors	0.617	1.468



12. Result

The relationship between CGPA and salary package should not be interpreted as direct causation. The observed correlation may partly arise because high-CGPA students also tend to possess other characteristics, such as internship experience, technical skills, and strong project profiles.

The comparison between the simple and multiple regression models demonstrates why additional variables should be included in the analysis. If the multiple regression model performs substantially better than the CGPA-only model, it indicates that salary depends on several factors rather than CGPA alone.

Correlation analysis, subgroup analysis, visualization, and multiple regression provide a better way to identify genuine predictive relationships and detect potentially misleading correlations.

13. Conclusion

The reported relationships are not sufficient to claim that CGPA causes higher salary packages.

Although CGPA may be positively associated with salary, factors such as internship experience, technical skills, projects, communication skills, and department may influence the relationship.

Hence, the university should not conclude that increasing CGPA alone will automatically increase salary. A comprehensive analysis using correlation matrices, scatter plots, subgroup analysis, and multiple regression should be conducted before making such a conclusion.

Final Conclusion: Correlation does not imply causation.

Time-Dependent Correlation Analysis – Advertising and Sales

1. Problem Statement

A retail company analyzes five years of monthly sales data and finds a very strong correlation of 0.93 between advertising expenditure and sales. Based on this relationship, management concludes that advertising is responsible for the increase in sales and that increasing the advertising budget will proportionally increase revenue.

However, several other factors changed during the same period, including the opening of new stores, product price increases, customer-base expansion, seasonal sales patterns, and the introduction of online sales.

The purpose of this analysis is to determine whether the correlation of 0.93 provides sufficient evidence that advertising caused the increase in sales.

2. Interpretation of the Correlation

A correlation coefficient of 0.93 indicates a very strong positive relationship between advertising expenditure and sales.

This means that months or periods with higher advertising expenditure generally tend to have higher sales.

However, this correlation does not prove that advertising caused the increase in sales.

Both variables have steadily increased over the five-year period. Therefore, the high correlation may partly result from a common time trend.

For example:

Time → Advertising increases

and

Time → Sales increases

This can create a strong correlation even when the direct causal effect of advertising is smaller than expected.

3. Why Correlation 0.93 Is Not Sufficient

The correlation of 0.93 is not enough to establish causation because several other factors may have contributed to the increase in sales.

The important issue is spurious correlation. Two variables may appear strongly related because both are increasing over time or because they are affected by other variables.

Therefore High correlation does not necessarily mean high causal impact.

The management cannot conclude that increasing advertising by a particular percentage will automatically increase revenue by the same percentage.

4. Confounding Factors

4.1 New Store Openings

The company opened new stores during the five-year period. More stores provide more opportunities to sell products, which can increase total sales independently of advertising.

4.2 Product Price Increases

If product prices increased, total revenue could increase even if the number of products sold remained unchanged.

Therefore, revenue growth may not be entirely due to advertising.

4.3 Customer Base Expansion

The number of customers increased during the period. A larger customer base can naturally produce higher sales.

4.4 Seasonality

Sales may increase during festivals, holidays, weekends, or particular months every year.

For example, sales may be consistently higher during Diwali or Christmas. If advertising expenditure also increases during these periods, the correlation between advertising and sales may become stronger.

4.5 Introduction of Online Sales

Online sales were introduced during the third year. This may have produced a significant increase in sales that is unrelated to advertising expenditure.

4.6 Long-Term Advertising Trend

Advertising expenditure increased almost every year regardless of short-term sales performance. Therefore, the relationship may primarily reflect the long-term upward trend rather than a direct response of sales to advertising.

5. Patterns That Should Be Investigated

The company should examine the following patterns separately.

A. Time Trend

Plot advertising expenditure and sales against time.

This will show whether both variables are simply increasing over the five-year period.

B. Seasonal Pattern

Analyze monthly sales separately to identify recurring seasonal peaks.

The company should determine whether advertising has a stronger effect during particular seasons.

C. Store Expansion

Compare sales before and after new stores were opened.

This helps determine whether increased sales are associated with store expansion rather than advertising.

D. Online Sales

Analyze sales before and after online sales were introduced in the third year.

This can identify whether the online channel produced a major structural change.

E. Price Changes

Separate the effect of price increases from actual changes in sales volume.

F. Customer Growth

Examine whether sales increased mainly because the number of customers increased.

6. Analytical Workflow

Step 1 – Data Visualization

Create monthly line charts showing:

- Advertising expenditure
- Sales/revenue
- Number of stores
- Customer base
- Product prices
- Online sales

This provides an initial understanding of the trends.

Step 2 – Correlation Analysis

Calculate the correlation between advertising and sales.

The observed correlation is:

$$r = 0.93$$

This confirms a strong positive association but does not establish causation.

Step 3 – Time-Series Analysis

Analyze the data over time to identify:

- Long-term trends
- Seasonal patterns
- Sudden changes
- Changes after online-sales introduction
- Changes after new-store openings

The correlation should also be examined after removing or controlling for the time trend.

Step 4 – Subgroup Analysis

Separate the data into meaningful groups such as:

- Before vs. after online sales
- Before vs. after store expansion
- High-season vs. low-season months
- Different store locations
- Different product categories

This helps determine whether the advertising-sales relationship remains consistent across groups.

Step 5 – Multiple Regression

A multiple regression model can be used to control for other factors:

$$\text{Sales} = \beta_0 + \beta_1(\text{Advertising}) + \beta_2(\text{Stores}) + \beta_3(\text{Price}) + \beta_4(\text{Customer Base}) + \beta_5(\text{Online Sales}) + \text{Seasonal Factors} + \text{Error}$$

Here, β_1 represents the estimated effect of advertising after controlling for the other variables.

Step 6 – Statistical Testing

Test whether the advertising coefficient is statistically significant.

Useful measures include:

- p-value
- Confidence interval
- R^2 / Adjusted R^2
- Regression coefficients
- Residual analysis

If advertising remains statistically significant after controlling for the other factors, there is stronger evidence that advertising has a genuine predictive relationship with sales.

7. Critical Evaluation of Management's Conclusion

Management states: "Advertising is responsible for the increase in sales, so increasing the advertising budget will proportionally increase revenue."

This conclusion is too strong.

The correlation of 0.93 only indicates a strong association. It does not establish that advertising is responsible for all or most of the increase in sales.

For example, sales may have increased because the company:

Opened new stores + increased prices + gained customers + introduced online sales while advertising expenditure also increased.

Therefore, simply increasing advertising may not produce a proportional increase in revenue.

CODE

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm

# -----
# 1. Create 5 years monthly data
# -----

np.random.seed(42)

months = pd.date_range("2021-01-01", periods=60, freq="MS")

# Advertising expenditure
advertising = np.linspace(20, 80, 60) + np.random.normal(0, 4, 60)

# Number of stores
stores = np.concatenate([
    np.full(20, 5),
    np.full(20, 8),
    np.full(20, 12)
])

# Customer base
customers = np.linspace(1000, 3000, 60) + np.random.normal(0, 100, 60)

# Product price
price = np.linspace(100, 130, 60)

# Online sales introduced from third year
online_sales = np.zeros(60)
online_sales[24:] = np.linspace(10, 50, 36)

# Seasonal effect
season = np.array([
    1.00, 0.95, 1.00, 1.05, 1.00, 1.05,
    1.00, 1.00, 1.05, 1.15, 1.30, 1.20
] * 5)

# Sales
sales = (
    100
    + 1.2 * advertising
    + 8 * stores
    + 0.03 * customers
    + 0.5 * price
    + 1.5 * online_sales
    + 40 * season
```

```

+ np.random.normal(0, 10, 60)
)

data = pd.DataFrame({
    "Month": months,
    "Advertising": advertising,
    "Sales": sales,
    "Stores": stores,
    "Customers": customers,
    "Price": price,
    "Online_Sales": online_sales,
    "Season": season
})

# -----
# 2. Display dataset
# -----

print("First 10 rows:")
print(data.head(10))

# -----
# 3. Correlation
# -----

correlation = data["Advertising"].corr(data["Sales"])

print("\nCorrelation between Advertising and Sales:")
print(round(correlation, 2))

# -----
# 4. Scatter Plot
# -----

plt.figure(figsize=(8, 5))
plt.scatter(data["Advertising"], data["Sales"], alpha=0.7)
plt.xlabel("Advertising Expenditure")
plt.ylabel("Sales")
plt.title("Advertising Expenditure vs Sales")
plt.grid(True)
plt.show()

# -----
# 5. Time Trend
# -----

plt.figure(figsize=(10, 5))

plt.plot(data["Month"], data["Advertising"], label="Advertising")
plt.plot(data["Month"], data["Sales"], label="Sales")

```



```

plt.xlabel("Year")
plt.ylabel("Value")
plt.title("Advertising and Sales Over Time")
plt.legend()
plt.grid(True)
plt.show()

# -----
# 6. Monthly Seasonal Pattern
# -----

data["Month_Number"] = data["Month"].dt.month

monthly_sales = data.groupby("Month_Number")["Sales"].mean()

plt.figure(figsize=(8, 5))
plt.plot(monthly_sales.index, monthly_sales.values, marker="o")
plt.xlabel("Month")
plt.ylabel("Average Sales")
plt.title("Monthly Seasonal Sales Pattern")
plt.xticks(range(1, 13))
plt.grid(True)
plt.show()

# -----
# 7. Before and After Online Sales
# -----

data["Online_Status"] = np.where(
    data["Online_Sales"] > 0,
    "After Online Sales",
    "Before Online Sales"
)

print("\nAverage Sales Before and After Online Sales:")
print(data.groupby("Online_Status")["Sales"].mean())

# -----
# 8. Multiple Regression
# -----

X = data[
    [
        "Advertising",
        "Stores",
        "Customers",
        "Price",
        "Online_Sales"
    ]
]

y = data["Sales"]

```

```
X = sm.add_constant(X)

model = sm.OLS(y, X).fit()

print("\nMultiple Regression Results:")
print(model.summary())

# -----
# 9. Advertising coefficient
# -----

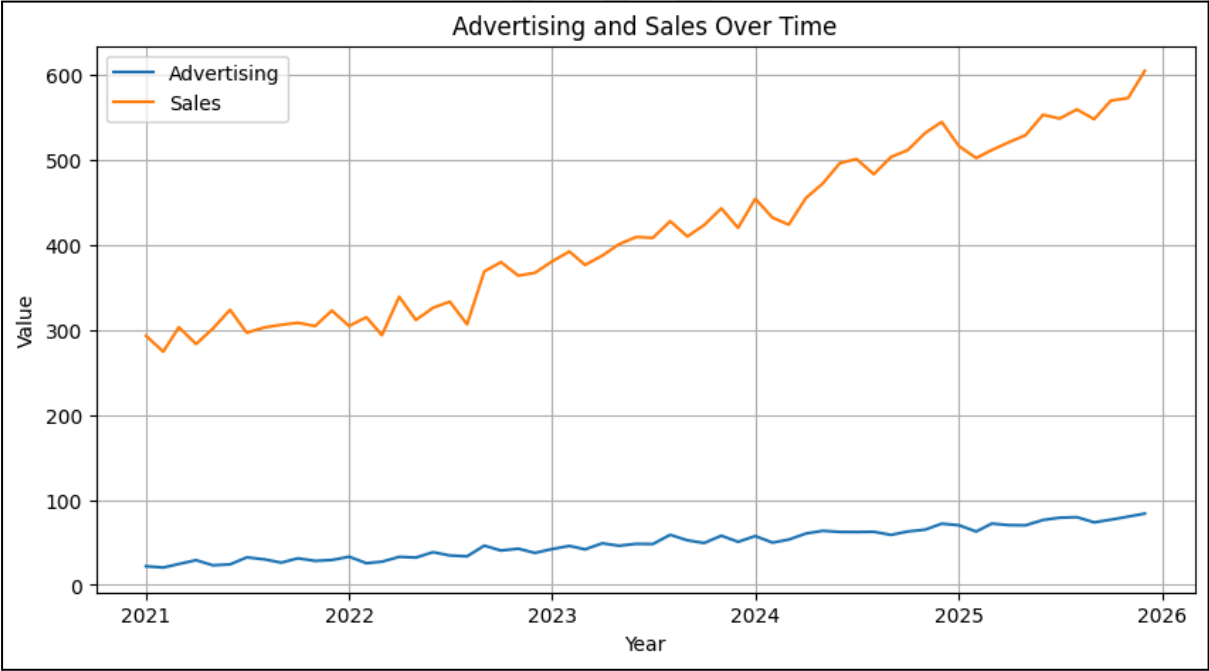
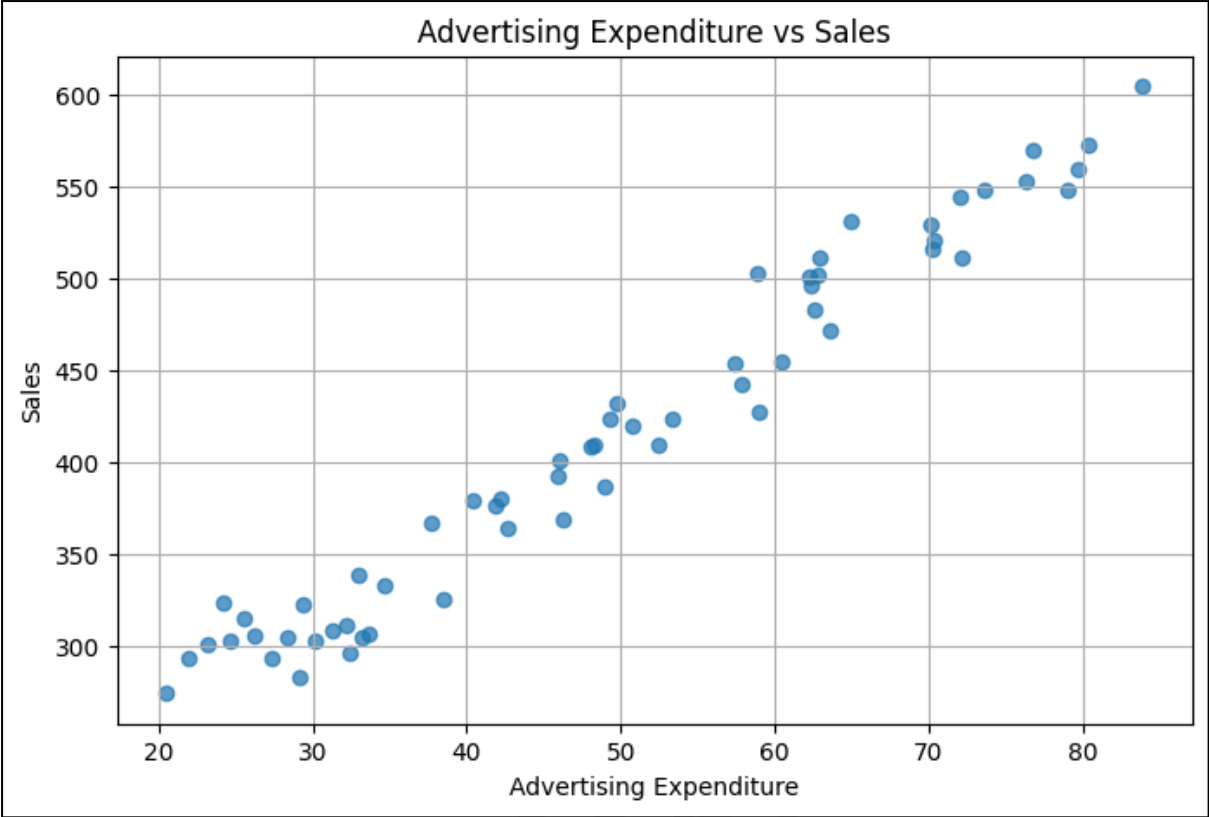
print("\nAdvertising Coefficient:")
print(model.params["Advertising"])

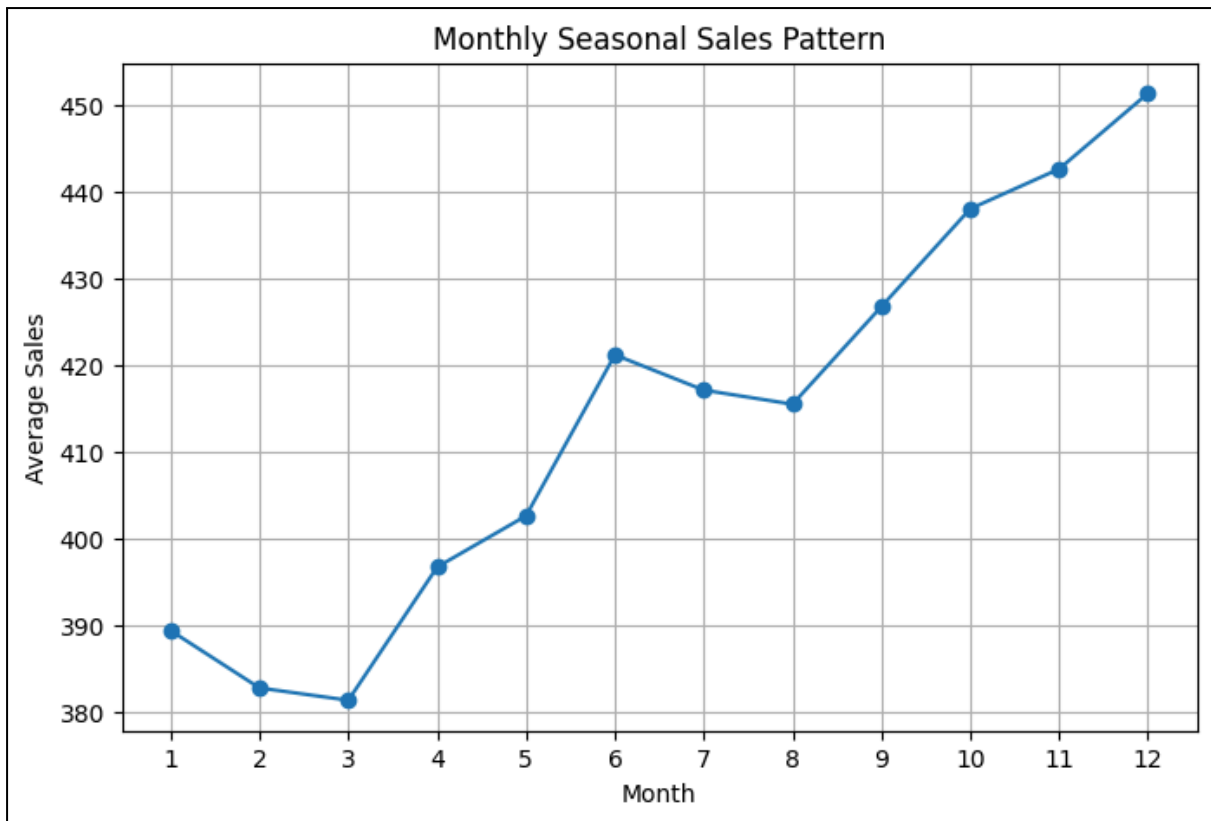
print("\nAdvertising p-value:")
print(model.pvalues["Advertising"])

# -----
# 10. Conclusion
# -----

if model.pvalues["Advertising"] < 0.05:
    print("\nAdvertising has a statistically significant relationship
with Sales.")
else:
    print("\nAdvertising is not statistically significant after
controlling for other variables.")
```

OUTPUT





Average Sales Before and After Online Sales:

Online_Status

After Online Sales 477.983449

Before Online Sales 317.605867

Name: Sales, dtype: float64

Multiple Regression Results:

OLS Regression Results

```

=====
Dep. Variable:      Sales      R-squared:      0.989
Model:              OLS       Adj. R-squared:  0.988
Method:             Least Squares   F-statistic:  956.8
Date:               Sat, 22 Aug 2026   Prob (F-statistic): 2.16e-51
Time:               05:59:36    Log-Likelihood: -223.03
No. Observations:   60         AIC: 458.1
Df Residuals:       54         BIC: 470.6
Df Model:           5
Covariance Type:    nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	370.1172	113.976	3.247	0.002	141.609	598.625
Advertising	1.5443	0.403	3.836	0.000	0.737	2.351
Stores	7.8999	1.452	5.440	0.000	4.988	10.812
Customers	0.0641	0.015	4.404	0.000	0.035	0.093
Price	-2.2082	1.267	-1.743	0.087	-4.748	0.332
Online_Sales	1.5272	0.322	4.750	0.000	0.883	2.172

```

=====
Omnibus:           3.239   Durbin-Watson:      2.073
Prob(Omnibus):     0.198   Jarque-Bera (JB):    2.390
Skew:              0.330   Prob(JB):            0.303
Kurtosis:          2.280   Cond. No.            1.76e+05
=====

```

8. Result

The analysis indicates that advertising expenditure and sales have a very strong positive correlation of 0.93. However, the relationship may be influenced by time trends, seasonality, store expansion, price changes, customer growth, and online sales.

Therefore, the correlation should not be interpreted as direct evidence of causation.

9. Conclusion

A correlation of 0.93 provides evidence of a strong association between advertising and sales, but it is not sufficient to prove that advertising caused the increase in sales.

The company should first control for time trends, seasonality, store expansion, prices, customer growth, and online sales.

A combination of visualization, time-series analysis, subgroup analysis, and multiple regression with statistical testing should be used to determine whether advertising has a genuine effect on sales.

Thus:

Correlation $\neq$ Causation

and management should not assume that increasing advertising will automatically and proportionally increase revenue.